

Resource Augmentation Analysis of the Greedy Algorithm for the Online Transportation Problem

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Introduction Forecast

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- **Result:** Constant competitiveness

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Goal: Minimize aggregate distance

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Algorithm: Greedy Algorithm

Motivation

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- OTP Central Problem

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 - Ride-hailing platforms

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 - **Theory:** Exponential competitiveness
 - **Practice:** Strong performance
- Closing the Gap
- **Q:** How can we rigorously explain Greedy's strong practical performance?

Twist: Resource Augmentation

Result: Constant Competitiveness

Body Outline

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① The Online Transportation Problem

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- 1 The Online Transportation Problem
- 2 Related Work

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- 3 Results and Key Insights

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- ③ Results and Key Insights
- ④ Summary

The Online Transportation Problem

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Setting

- $\mathcal{M}$: Metric Space

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- Minimize the cost of the matching

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Greedy Algorithm

- Assigns requests to nearest available server

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- Optimal matching still has capacities a_j

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Research Question

- **Q.** What is the competitive ratio of GREEDY under resource augmentation $k \geq 3$?

Related Work

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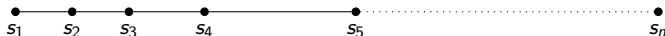
- $k = 1$: [1] GREEDY is $(2^m - 1)$ -competitive
- $k = 2$: [2] GREEDY is $\Theta(\min(m, \log C))$ -competitive
- $k \geq 3$: GREEDY is $\left(1 + \frac{2}{k-2}\right)$ -competitive

$k = 1$ GREEDY Competitiveness

Consider the line metric and suppose there are single servers at $s_1 = -1 - \varepsilon$ (for some $\varepsilon > 0$) and $s_i = 2^{i-1} - 1$ (for $2 \leq i \leq n$) as shown below. Then, we give a request sequence where GREEDY has a competitive ratio of $2^n - 1$:

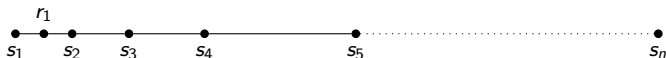
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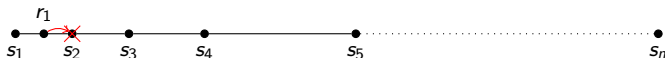
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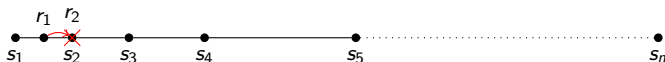
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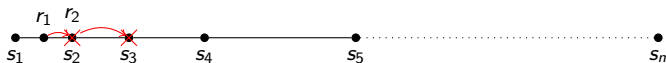
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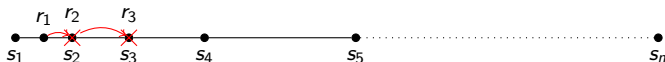
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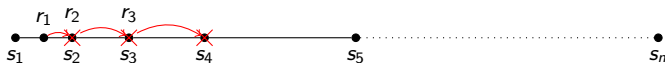
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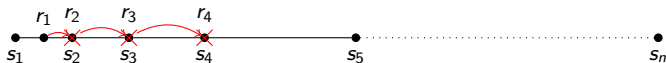
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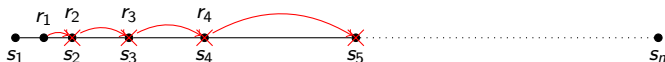
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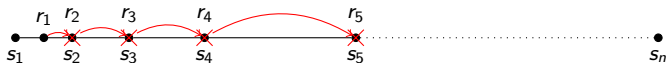
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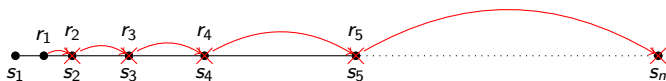
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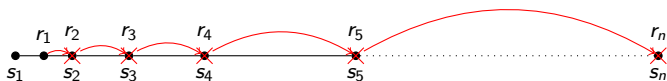
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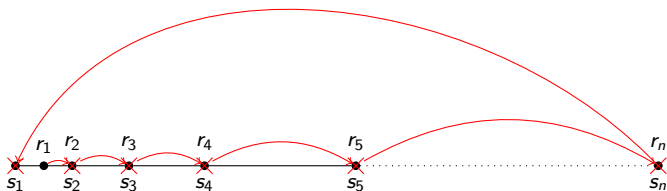
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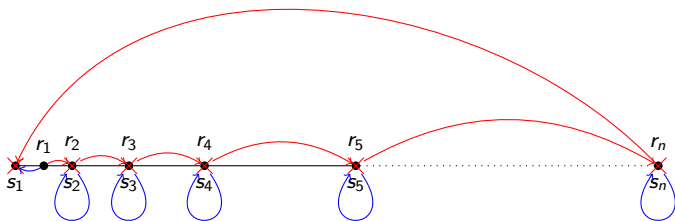
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Initial Assumption

Assumption: $a_j = 1, c_j = k$ for all server sites

Response Graph

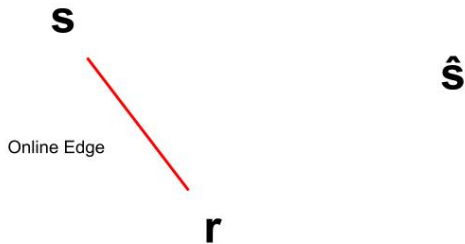
Response Graph

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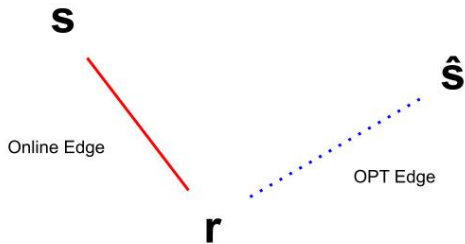
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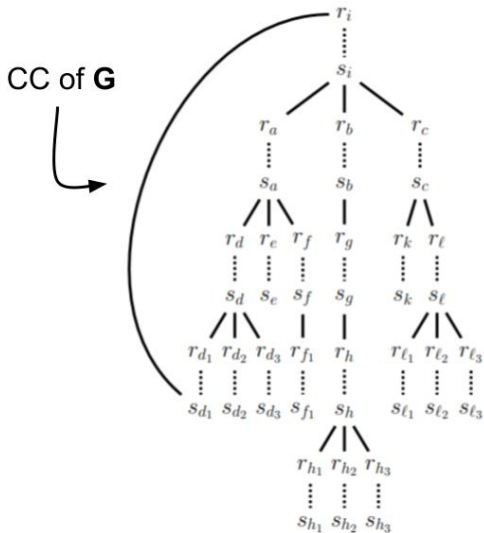
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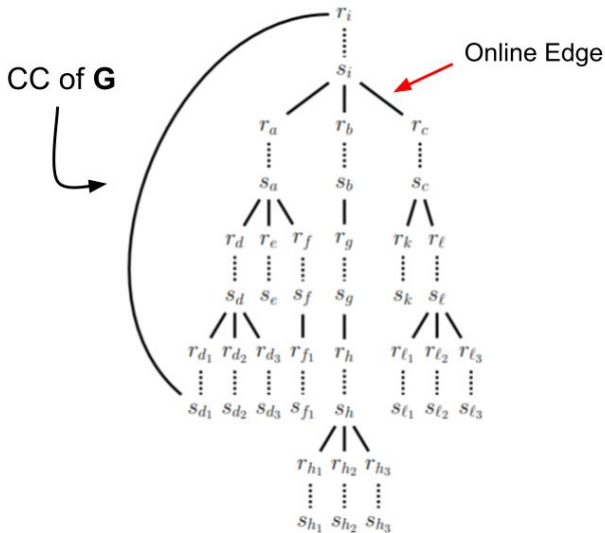
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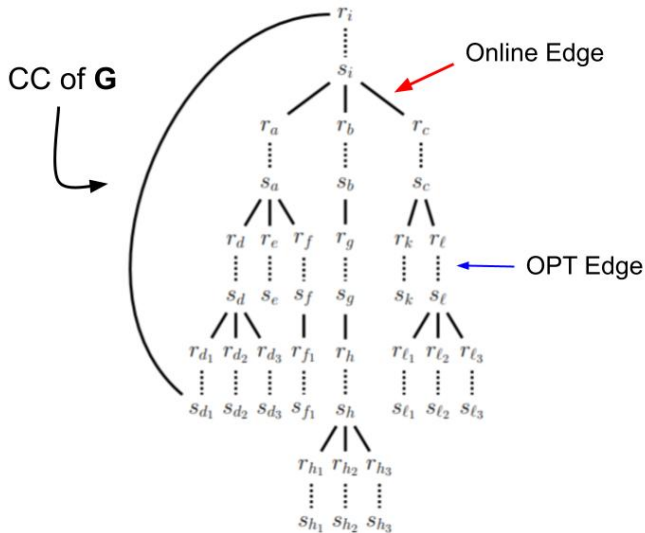
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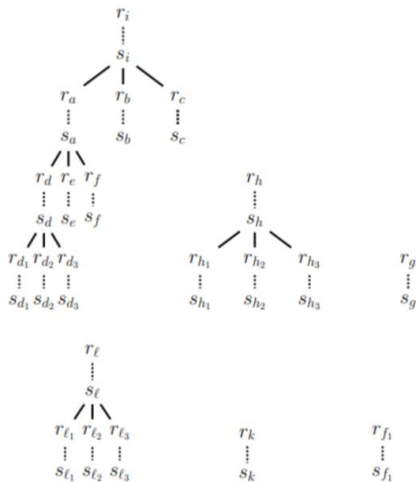
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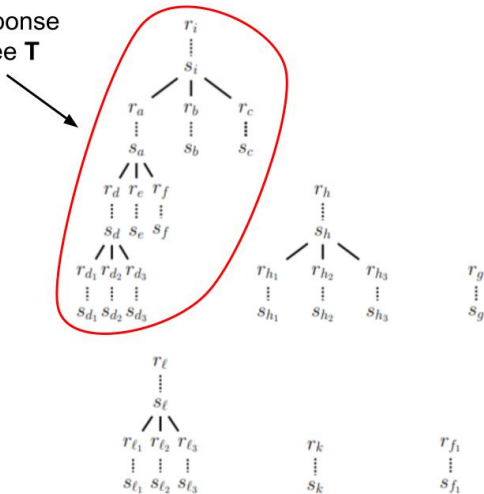


Response Trees



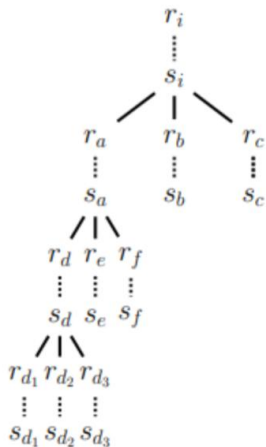
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Response
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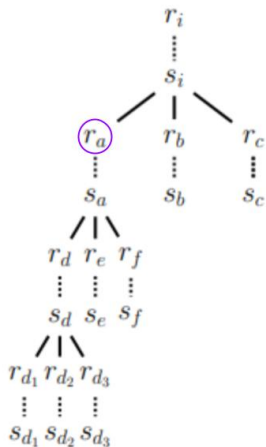


Response Trees

Want: Bound **online** edges by optimal edges

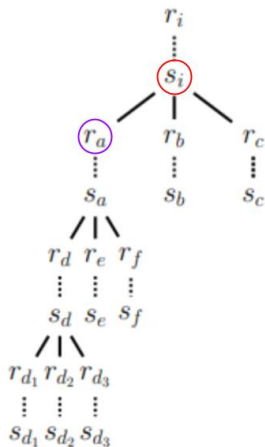
Online Edge $\leq$ Leaf Distance

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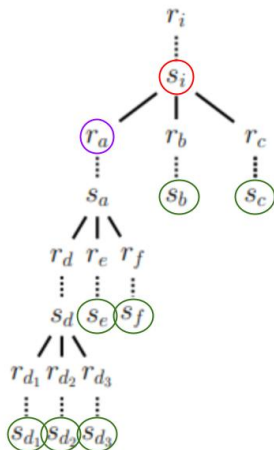
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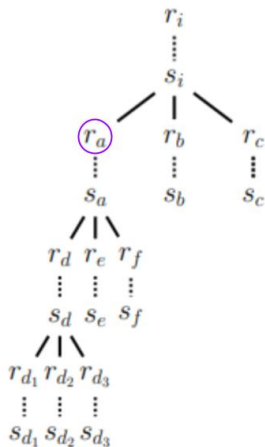


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Leaf Distance: Shortest path from request to leaf-server in $\mathcal{T}$

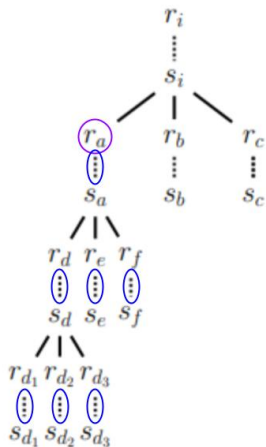
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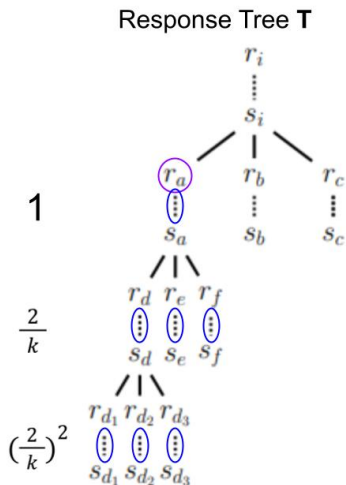
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Weighted Tree Cost: Weighted sum of optimal edges below request in $\mathcal{T}$. Edge weights decrease **geometrically** by $\frac{2}{k}$.

Leaf Distance $\leq$ Weighted Tree Cost



Aggregating Edges

OPT Edge Weights: $1 + \frac{2}{k} + \left(\frac{2}{k}\right)^2 + \dots = \boxed{1 + \frac{2}{k-2}}$

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Remove Assumption: Allow all $a_j \geq 1$, and let $c_j = k \cdot a_j$.

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- $k \geq 3$ RA offsets GREEDY's limitations

Bibliography



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Online weighted matching.

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Questions?

